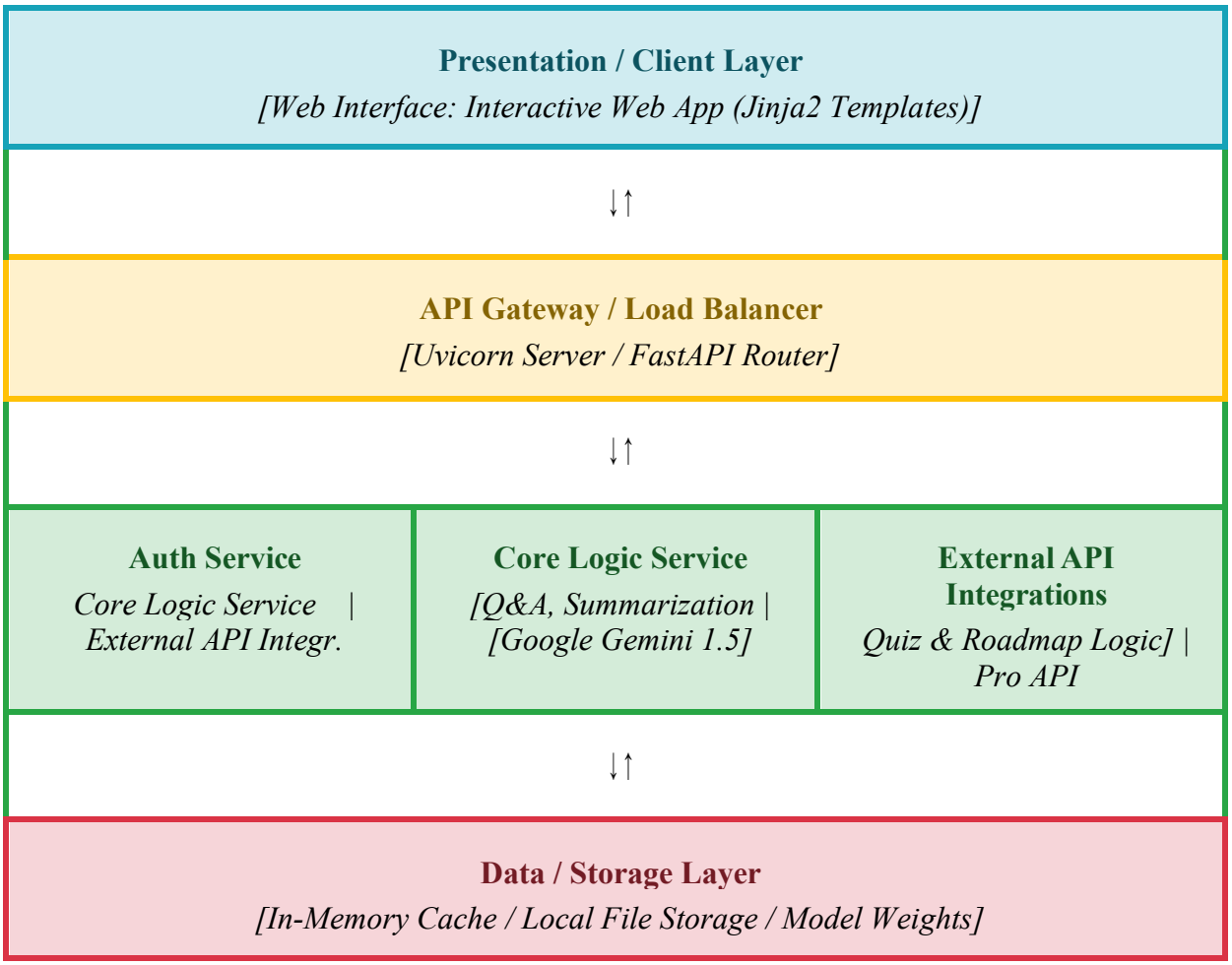
Solution Architecture

Date	02 October 2026
Team ID	SWTID-2026-3981
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides the user interface; accepts inputs (questions, topics, and passages) and displays AI-generated answers, quizzes, summaries, and learning roadmaps.	HTML, CSS, JavaScript, Jinja2 Templating
[API Gateway]	Receives client requests and passes them to the processing services through dynamic API routes (/qa, /explain, /quiz, /summarize, /learn/recommendations).	FastAPI, Uvicorn ASGI Server
[Microservices]	Handles AI capabilities, including prompt validation and query routing between the cloud API and the local offline inference model.	Python 3.10+, PyTorch, Transformers Library, LaMini-Flan-T5-783M Model
[Database]	Used to hold session status, local model files/weights, and response output buffers.	Stateless In-Memory Session Storage / Local File System